GDC Video Repair

Complete guide: installation, how it works, examples, troubleshooting

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Guide updated for the current app version

1. What is GDC Video Repair

GDC Video Repair fixes corrupted MP4/MOV video files — the most common case: the video card was pulled from the camera mid-recording, or the transfer to a computer was interrupted. It uses the reference-file technique, the same method used by well-known tools like Untrunc.

The problem, briefly

An MP4/MOV file has two essential parts: mdat (the raw video data, the actual frames) and moov (the "map" telling the player where each frame starts, which codec is used, the resolution). The camera writes mdat continuously, but only writes moov at the end, when you stop recording. If you pull the card mid-recording, moov is either completely missing or incomplete — the file becomes unreadable, even though the raw video data is often perfectly intact.

What it can do

- Repairs H.264 and H.265 (MP4/MOV) files with missing or corrupted moov.
- Partially recovers files truncated mid-recording — everything recoverable, up to the exact point of corruption.
- Tries a quick remux first (no reference file needed) for easy cases.
- Simple GUI, or command line for automated workflows.

What it doesn't do (yet)

- Single video track only — doesn't reconstruct audio tracks if completely missing.
- H.264 and H.265 only for reference-based reconstruction.
- Doesn't repair files where the raw video data (mdat) itself is corrupted at the content level, not just the header.

Note: The app is free and open-source (MIT license) — full source code is available on GitHub.

2. Installation

Required: FFmpeg

The app needs FFmpeg installed separately on the system — it doesn't come bundled.

```
# macOS
brew install ffmpeg

# Windows: download from ffmpeg.org and add to PATH
```

macOS

Download GDCVideoRepair.pkg from the Releases page. Double-click and follow the standard macOS installer.

Windows

Download GDCVideoRepair.exe from the Releases page. Double-click to run directly — no separate installation needed.

> **Note:** If the app can't find FFmpeg, check from Terminal/Command Prompt that running "ffmpeg -version" works — if not, FFmpeg isn't correctly installed or isn't in PATH.

3. How the repair works, step by step

Step 1 — quick remux

The app first tries a simple remux (no reference file) — resolves easy cases, where moov exists but something minor is off.

Step 2 — reference-based reconstruction

If step 1 isn't enough, the app:

- Analyzes the healthy reference file and extracts the exact codec configuration.
- Scans the corrupt file's raw video data byte by byte, identifying the real boundaries of each frame.
- Builds a new, fully valid moov, using the reference's configuration + the real frames found.
- Writes the final file, with the original video data untouched.

Choosing a good reference file

- Filmed with the exact same camera — ideally the same physical device.
- Same settings: resolution, framerate, codec (H.264 vs H.265).
- Content doesn't matter — any healthy clip works, even a few-second test.

4. Usage examples

Graphical interface

- Open the app.
- Choose the corrupt file.
- Choose the reference file (optional, but recommended).
- Choose where to save the repaired file.
- Click Repair.

Command line

```
GDCVideoRepair.exe --corrupt corrupt.mp4 --reference healthy.mp4 --output repaired.mp4
```

The reference file is optional — if the quick remux succeeds on its own, you don't need it.

5. Common issues and solutions

"Couldn't identify valid video samples"

- The corrupt file is likely damaged at the raw data level too, not just the header.
- Check if the original file has at least a few KB of recognizable data.

"Reference file doesn't have a valid video track"

- Make sure the reference file actually plays normally before using it.

Repaired file plays but the image is broken

- The codec or configuration in the reference doesn't exactly match the corrupt file.
- Try a different reference file, filmed with settings as close as possible to the corrupt one.

Quick remux takes a long time

- For very large files (4K/6K, many minutes), the initial remux can take a while — let it finish.

6. License

GDC Video Repair is MIT licensed — free, open-source, full source code available on GitHub. You can use, modify, and redistribute it freely.

The app uses FFmpeg (installed separately by the user, not bundled) as an external tool for quick remuxing and validation — it doesn't link directly against FFmpeg's libraries.